

AUTOSAR High-Level Design Document

Project: Smart Vehicle Control System

1. System Overview

The system provides vehicle sensor monitoring, communication, diagnostic services, and vehicle control functionality.

2. Software Components

The major software components are:

1. Sensor Manager
2. Communication Manager
3. Diagnostic Manager
4. Vehicle Control Manager

3. Interfaces and Dependencies

Sensor Manager collects and processes sensor data and communicates with the Communication Manager.

Communication Manager is responsible for vehicle network communication. It receives processed information from the Sensor Manager and sends information to the Vehicle Control Manager.

Diagnostic Manager handles diagnostic requests and responses and communicates with the Communication Manager.

4. Functional Flow

The functional flow of the system is:

Sensors ? Sensor Manager ? Communication Manager ? Vehicle Control Manager

Diagnostic requests received by the Diagnostic Manager are forwarded through the Communication Manager.

5. Important Signals

The important signals used by the system are:

- ? VehicleSpeed
- ? EngineTemperature
- ? BrakeStatus
- ? DiagnosticRequest